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Unit 09: Association and Correlation Analysis

CONTENTS

Objectives

After this lecture, you will be able to

- Understand the concept of frequent patterns and association rules.
- Learn how to calculate support and confidence.
- Understand the basic concepts of the Apriori algorithm.
- Learn the working of the Apriori algorithm.
- Understand the process of finding frequent patterns using FP-tree.
- Know the applications of association rule mining.

Introduction

Imagine that you are a sales manager, and you are talking to a customer who recently bought a PC and a digital camera from the store. What should you recommend to her next?Frequent patterns and association rules are the knowledge that you want to mine in such a scenario.The term "association mining" refers to the process of looking for frequent items in a data set. Typically, interesting connections and similarities between item sets in transactional and relational databases are discovered through regular mining. Frequent Mining, in a nutshell, shows which objects appear together in a transaction or relationship.

9.1 Basic Concepts

Frequent patterns are patterns (e.g., itemsets, or subsequences) that appear frequently in a data set.Frequent Pattern Mining is a Data Mining topic that aims to extract frequently occurring itemsets from a database. Frequent itemsets are linked to interesting trends in data, such as Association Rules, and play an important role in many Data Mining tasks.

For example, a set of items, such as milk and bread, that appear frequently together in a transaction data set is a frequent item-set.

A subsequence, such as buying first a PC, then a digital camera, and then a memory card, if it occurs frequently in a shopping history database, is a (frequent) sequential pattern.

< {Homepage} {Electronics} {Digital Cameras} {Canon Digital Camera} {Shopping Cart} {Order Confirmation} {Return to Shopping} >

Subgraphs, subtrees, and sublattices are examples of structural structures that can be combined with itemsets or subsequences to form a substructure. A (frequent) structural pattern is a substructure that appears regularly in a graph database. In mining associations, correlations, and many other interesting relationships among data, finding frequent patterns is critical.

Market Basket Analysis: A Motivating Example

Market basket analysis is a common example of frequent itemset mining. This method examines consumer purchasing patterns by identifying links between the various products that customers put in their "shopping baskets."The discovery of these associations can help retailers develop marketing strategies by gaining insight into which items are frequently purchased together by customers.Customers buying milk, for example, are they more likely to buy bread (and what kind of bread) on the same trip to the supermarket? This data will help retailers manage their shelf space and do selective promotions, which can lead to increased sales.

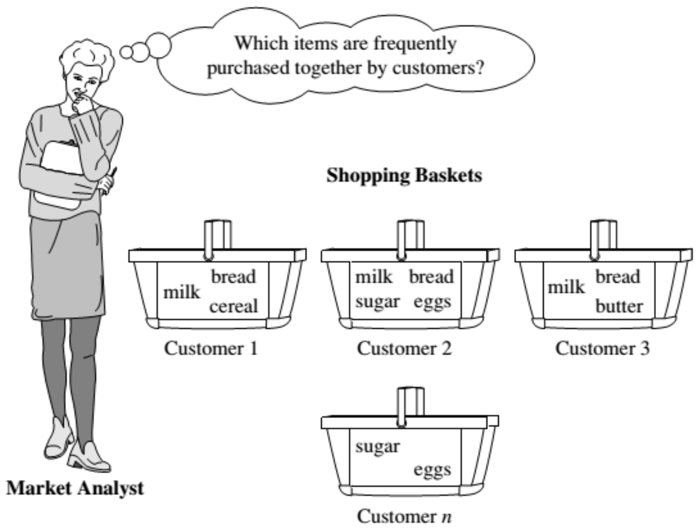

Figure 1: Market Basket Analysis

9.2 How does Association Rule Learning work?

Association rules are "if-then" statements that display the likelihood of relationships between data items in large data sets in a variety of databases.

Rule form

Antecedent → Consequent [support, confidence]

(support and confidence are user-defined measures of interestingness)

If A then B, for example, association rule learning is based on the principle of If and Else Statements.

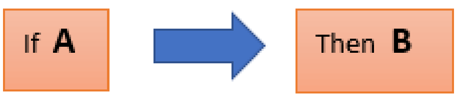

Figure 2: Association Rule

The If aspect is referred to as the antecedent, and the Then statement is referred to as the Consequent. Single cardinality refers to relationships in which we may discover an interaction or relationship between two objects. It's all about making laws, and as the number of things grows, so does cardinality.

age(x, “30..39”) ^ income(x, “42..48K”) →buys(x, “car”) [1%,75%]

Frequent Patterns and Association Rules

If we think of the universe as the set of items available at the store, then each item has a Boolean variable representing the presence or absence of that item. Each basket can then be represented by a Boolean vector of values assigned to these variables. The Boolean vectors can be analyzed for buying patterns that reflect items that are frequently associated or purchased together. These patterns can be represented in the form of ASSOCIATION RULES.

We used association rules to find mistakes often occurring together while solving exercises.The purpose of looking for these associations is for the teacher to ponder and, may be, toreview the course material or emphasize subtleties while explaining concepts to students. Thus, it makes sense to have a support that is not too low.

For example, the information that customers who purchase computers also tend to buy antivirus software at the same time is represented in the following association rule:

Computer=>antivirus_Software [support=2%, confidence=60%].

Support of 2% for Rule means that 2% of all the transactions under analysis show that computer and antivirus software are purchased together.A confidence of 60% means that 60% of the customers who purchased a computer also bought the software. There are several metrics for measuring the relationships between thousands of data objects. These figures are as follows:

Support

Confidence

Support

The frequency of A, or how often an object appears in the dataset, is called "support." It's the percentage of the transaction T that has the itemset X in it. If there are X datasets, the following can be written for transaction T:

Support(X)=Freq(X)/T

Confidence

The term "confidence" refers to how much the rule has been proven correct. Or, when the frequency of X is already known, how often the items X and Y appear together in the dataset. It's the ratio of the number of records that contain X to the number of transactions that contain X.

Confidence=Freq(X,Y)/Freq(X)

Table 1: DataItems along with Transaction ids

TID	Items
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

From the above table, {Milk, Diaper}=>{Beer} s= σ({Milk, Diaper, Beer}) $\div$ |T|

= 2/5

= 0.4

c= σ (Milk, Diaper, Beer) $\div$ (Milk, Diaper)

= 2/3

= 0.67

Rules that satisfy both a minimum support threshold (min_sup) and a minimum confidence threshold (min_conf) are called strong. A set of items is referred to as an itemset. An itemset that contains k items is a k-itemset. The set **{computer, antivirus software}** is a 2-itemset. The occurrence frequency of an itemset is the number of transactions that contain the itemset. This is also known, simply, as the **frequency, support count, or count** of the itemset.

In general, association rule mining can be viewed as a two-step process:

- ***Find all frequent itemsets*:** By definition, each of these itemsets will occur at least as frequently as a predetermined minimum support count, min sup.
- ***Generate strong association rules from the frequent itemsets*:** By definition, these rules must satisfy minimum support and minimum confidence.

It could be useful for the OurVideoStore manager to know what movies are often rented together or if there is a relationship between renting a certain type of movie and buying popcorn or pop. The discovered association rules are of the form: P→Q [s, c], where P and Q are conjunctions of attribute value-pairs, and s (for support) is the probability that P and Q appear together in a transaction and c (for confidence) is the conditional probability that Q appears in a transaction when P is present. For example, the hypothetic association rule

Rent Type (X, “game”) ^ Age(X,“13-19”) → Buys(X, “pop”) [s=2% ,c=55%]

would indicate that 2% of the transactions considered are of customers aged between 13 and 19 who are renting a game and buying pop and that there is a certainty of 55% that teenage customers, who rent a game, also buy pop.

Why Frequent Itemset Mining?

Because of its many applications in mining association rules, correlations, and graph pattern constraints that are focused on frequent patterns, sequential patterns, and many other data mining tasks, frequent itemset or pattern mining is widely used.

9.3 The Apriori Algorithm: Basics

The following are the key concepts used in context to apriori algorithm :

- **Frequent Itemsets**: The sets of an item that has minimum support
- **Apriori Property**: Any subset of frequent item-set must be frequent.
- **Join Operation**: To find Lk, a set of candidate k-item-sets is generated by joining Lk-1 with itself.

Apriori Algorithm

The Apriori algorithm was the first algorithm for frequent itemset mining to be proposed. R Agarwal and R Srikant improved it later, and it became known as Apriori. To reduce the search space, this algorithm uses two steps: "join" and "prune." It is an iterative method for identifying the most frequent itemsets.

Apriori says:

The probability that item I is not frequent is if:

- P(I) < minimum support threshold, then I is not frequent.
- P (I+A) < minimum support threshold, then I+A is not frequent, where A also belongs to itemset.
- If an itemset set has a value less than minimum support then all of its supersets will also fall below min support, and thus can be ignored. This property is called the Antimonotone property.

The Apriori Algorithm for data mining includes the following steps:

1. **Join Step:** By joining each item with itself, this move generates (K+1) itemset from K-itemsets.
2. **Prune Step:** This step counts all of the items in the database. If a candidate item fails to fulfill the minimum support requirements, it is classified as infrequent and hence withdrawn. This step aims to reduce the size of the candidate itemsets.

Steps In Apriori

The apriori algorithm is a series of steps that must be followed to find the most frequent itemset in a database. This data mining technique repeats the join and prune steps until the most frequently occurring itemset is found. The issue specifies a minimum assistance threshold, or the user assumes it.

1) Each object is treated as a 1-itemsets candidate in the first iteration of the algorithm. Each item's occurrences will be counted by the algorithm.

2) Set a minimum level of support, min sup. The set of 1 – itemsets whose occurrence meets the minimum sup requirement is determined. Only those candidates with a score greater than or equal to min sup are advanced to the next iteration, while the rest are pruned.

3) Next, min sup is used to find 2-itemset frequent itemsets. The 2-itemset is formed in the join phase by forming a group of 2 by combining items with itself.

4) The min-sup threshold value is used to prune the 2-itemset candidates. The table will now have two –itemsets, one with min-sup and the other with just min-sup.

5) Using the join and prune step, the next iteration will create three –itemsets. This iteration will use the antimonotone property, which means that the subsets of 3-itemsets, i.e. the two –itemset subsets of each category, will fall into min sup. The superset will be frequent if all 2-itemset subsets are frequent, otherwise, it will be pruned.

6) Making 4-itemset by joining a 3-itemset with itself and pruning if its subset does not meet the min sup criteria would be the next move. When the most frequent itemset is reached, the algorithm is terminated.

Table 2: Itemsets

Transaction	List of items
T1	I1,I2,I3
T2	I2,I3,I4
T3	I4,I5
T4	I1,I2,I4
T5	I1,I2,I3,I5
T6	I1,I2,I3,I4

Example of Apriori: Support threshold=50%, Confidence= 60%

Solution:

Support threshold=50% => 0.5*6= 3 => min_sup=3

1. Frequency of each item.

Table 3: Items With Frequency Count

Item	Count
I1	4
I2	5
I3	4
I4	4
I5	2

2. Prune Step: Table 3 shows that the I5 item does not meet min_sup=3, thus it is deleted, only I1, I2, I3, I4 meet min_sup count.
3. Step 2: Form a 2-itemset. Find the occurrences of 2-itemset in Table 2.

Table 4: Two Itemset Data

Item	Count
I1,I2	4
I1,I3	3
I1,I4	2
I2,I3	4
I2,I4	3
I3,I4	2

4. Prune Step: Table 4reveals that item sets I1, I4, and I3, I4 do not follow min sup, so they are deleted.

Table 5: Frequent Two Itemset data

Item	Count
I1,I2	4
I1,I3	3
I2,I3	4
I2,I4	3

5. Join and Prune Step: join and prune Form a three-itemset. Find the occurrences of the 3-itemset in Table 2. Find the 2-itemset subsets that endorse min sup from Table 5.

We can see that for itemset{ I1, I2, I3} subsets, TABLE-5 contains {I1, I2}, {I3, I2},{I1, I3} indicating that {I1, I2, I3 } is frequent.

Table 6: Three Itemset Data

Item
I1,I2,I3
I1,I2,I4
I1,I3,I4

Only {I1, I2, I3} is frequent.

6. **Generate Association Rules:** From the frequent itemset discovered above the association could be:

{I1, I2} => {I3}

Confidence = support {I1, I2, I3} / support {I1, I2} = (3/ 4)* 100 = 75%

{I1, I3} => {I2}

Confidence = support {I1, I2, I3} / support {I1, I3} = (3/ 3)* 100 = 100%

{I2, I3} => {I1}

Confidence = support {I1, I2, I3} / support {I2, I3} = (3/ 4)* 100 = 75%

{I1} => {I2, I3}

Confidence = support {I1, I2, I3} / support {I1} = (3/ 4)* 100 = 75%

{I2} => {I1, I3}

Confidence = support {I1, I2, I3} / support {I2 = (3/ 5)* 100 = 60%

{I3} => {I1, I2}

Confidence = support {I1, I2, I3} / support {I3} = (3/ 4)* 100 = 75%

This shows that all the above association rules are strong if minimum confidence threshold is 60%.

Advantages

- The algorithm is simple to comprehend.
- On large itemets in large databases, the join and prune steps are simple to implement.

Disadvantages

- If the itemsets are wide and the minimum support is held low, it necessitates a lot of computation.
- The database as a whole must be scanned.

9.4 FP Growth Algorithm

The FP growth algorithm is a step forward from the apriori algorithm. Without candidate generation, the FP growth algorithm is used to find frequent itemsets in a transaction database. In frequent pattern trees or FPtrees, FP growth represents frequent items.

The relationship between the itemsets will be maintained by this tree structure. Using one frequent item, the database is fragmented. The broken portion is referred to as a "pattern fragment." These fragmented patterns' itemsets are examined. As a result, the search for frequently occurring itemsets is significantly reduced using this approach.

FP Tree

The Frequent Pattern Tree is a tree-like structure created from the database's initial itemsets. The FP tree aims to find the most frequent pattern. Each object in the itemset is represented by a node in the FP tree.Null is represented by the root node, while itemsets are represented by the lower nodes. When forming the tree, the connection of the nodes with the lower nodes, that is, the itemsets with the other itemsets is retained.

Steps In frequent pattern algorithm

We will find the frequent pattern using the frequent pattern growth method without having to generate candidates.

1) The first step is to search the database for instances of the itemsets. This move is identical to Apriori's first step. Help count or frequency of 1-itemset refers to the number of 1-itemsets in the database.
2) The FP tree is built in the second stage. To do so, start by making the tree's root. Null is used to represent the root.
3) The following move is to re-scan the database and look over the transactions. Examine the first transaction to determine the itemset contained therein. The highest-counting itemset is at the top, followed by the next-lowest-counting itemset, and so on. It means that the tree's branch is made up of transaction itemsets arranged in descending order of count.
4) The database's next transaction is analyzed. The itemsets are sorted by count in ascending order. This transaction branch will share a common prefix to the root if any itemset of this transaction is already present in another branch (for example, in the first transaction).This implies that in this transaction, the common itemset is connected to the new node of another itemset.
5) In addition, as transactions occur, the count of the itemset is increased. As nodes are generated and connected according to transactions, the count of both the common node and new node increases by one.
6) The next move is to mine the FP Tree that has been developed. The lowest node, as well as the relations between the lowest nodes, are analyzed first. The frequency pattern length 1 is represented by the lowest node. The conditional pattern base is a sub-database that contains prefix paths in the FP tree that start at the lowest node (suffix).
7) Create a Conditional FP Tree based on the number of itemsets in the route. The Conditional FP Tree considers the itemsets that meet the threshold support.
8) The Conditional FP Tree generates Frequent Patterns.

Example Of FP-Growth Algorithm

Support threshold=50%, Confidence= 60%

Table 7: Purchased Itemset

Transaction	List of items
T1	I1,I2,I3
T2	I2,I3,I4
T3	I4,I5
T4	I1,I2,I4
T5	I1,I2,I3,I5

T6	I1,I2,I3,I4

Support threshold=50% => 0.5*6= 3 => min_sup=3

1. Count of each item.

Table 8: Items with frequency Count

Item	Count
I1	4
I2	5
I3	4
I4	4
I5	2

2. Sort the itemset in descending order.

Table 9: Itemset in descending order of their frequency

Item	Count
I2	5
I1	4
I3	4
I4	4

3. Build FP tree.

a) Considering the root node null.

b) The first scan of Transaction T1: I1, I2, I3 contains three items {I1:1}, {I2:1}, {I3:1}, where I2 is linked as a child to root, I1 is linked to I2, and I3 is linked to I1.

c) T2: I2, I3, I4 contains I2, I3, and I4, where I2 is linked to root, I3 is linked to I2, and I4 is linked to I3. But this branch would share the I2 node as common as it is already used in T1.

d) Increment the count of I2 by 1 and I3 is linked as a child to I2, I4 is linked as a child to I3. The count is {I2:2}, {I3:1}, {I4:1}.

e) T3: I4, I5. Similarly, a new branch with I5 is linked to I4 as a child is created.

f) T4: I1, I2, I4. The sequence will be I2, I1, and I4. I2 is already linked to the root node, hence it will be incremented by 1. Similarly I1 will be incremented by 1 as it is already linked with I2 in T1, thus {I2:3}, {I1:2}, {I4:1}.

g) T5:I1, I2, I3, I5. The sequence will be I2, I1, I3, and I5. Thus {I2:4}, {I1:3}, {I3:2}, {I5:1}.

h) T6: I1, I2, I3, I4. The sequence will be I2, I1, I3, and I4. Thus {I2:5}, {I1:4}, {I3:3}, {I4 1}.

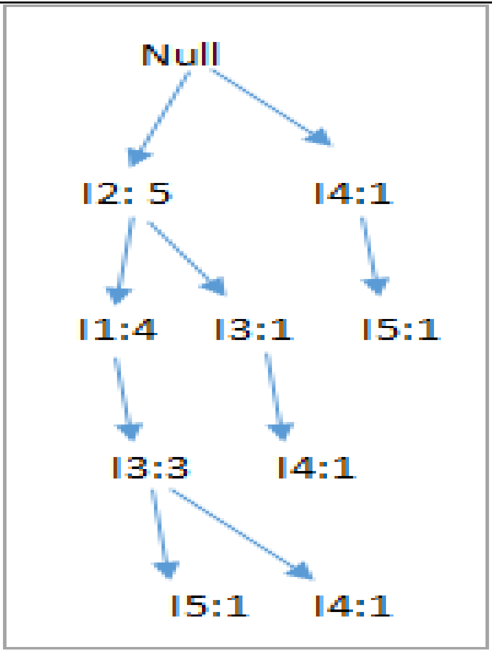

Figure 3: FP tree generation

Mining of FP-tree is summarized below:

1. The lowest node item I5 is not considered as it does not have a min support count, hence it is deleted.
2. The next lower node is I4. I4 occurs in 2 branches , {I2,I1,I3:,I41},{I2,I3,I4:1}. Therefore considering I4 as suffix the prefix paths will be {I2, I1, I3:1}, {I2, I3: 1}. This forms the conditional pattern base.
3. The conditional pattern base is considered a transaction database, an FP-tree is constructed. This will contain {I2:2, I3:2}, I1 is not considered as it does not meet the min support count.
4. This path will generate all combinations of frequent patterns : {I2,I4:2},{I3,I4:2},{I2,I3,I4:2}
5. For I3, the prefix path would be: {I2,I1:3},{I2:1}, this will generate a 2 node FP-tree : {I2:4, I1:3} and frequent patterns are generated: {I2,I3:4}, {I1:I3:3}, {I2,I1,I3:3}.
6. For I1, the prefix path would be: {I2:4} this will generate a single node FP-tree: {I2:4} and frequent patterns are generated: {I2, I1:4}.

Table 10: Process of Frequent Pattern Generation

Item	Conditional Pattern Base	Conditional FP-tree	Frequent Patterns Generated
I4	{I2,I1,I3:1},{I2,I3:1}	{I2:2, I3:2}	{I2,I4:2},{I3,I4:2},{I2,I3,I4:2}
I3	{I2,I1:3},{I2:1}	{I2:4, I1:3}	{I2,I3:4}, {I1:I3:3}, {I2,I1,I3:3}
I1	{I2:4}	{I2:4}	{I2,I1:4}

The Benefits of the FP Growth Algorithm

1. As compared to Apriori, which scans the transactions for each iteration, this algorithm only needs to scan the database twice.
2. This algorithm does not perform item pairing, which makes it faster.
3. The database is kept in memory in a compressed form.

4. It can be used to mine both long and short regular patterns and is both productive and scalable.

The FP-Growth Algorithm's Drawbacks

1. The FP Tree is more time-consuming and difficult to build than the Apriori.
2. It could be costly.
3. The algorithm can not fit in shared memory if the database is massive.

Discuss how the discovery of different patterns through different data miningalgorithms and visualisation techniques suggest you a simple pedagogical policy?

9.5 Applications of Association Rule Learning

It can be used in a variety of machine learning and data mining applications. The following are some of the most common uses of association rule learning:

Analysis of the Market Basket: One of the most well-known examples and implementations of association rule mining is this. Big retailers often employ this technique to evaluate the relationship between items.

Medical Diagnosis: Patients may be cured quickly using association guidelines, as they assist in determining the likelihood of infection with a specific condition.

Protein Sequence: The rules of association aid in the synthesis of artificial proteins.

It's also used for **Catalog Design, Loss-leader Analysis**, and a variety of other tasks.

Summary

- In selective marketing, decision analysis, and business management, the discovery of frequent trends, correlations, and correlation relationships among massive amounts of data is useful.
- A popular area of application is **market basket analysis**, which studies customers' buying habits by searching for itemsets that are frequently purchased together.
- The process of "association rule mining" entails first identifying frequent itemsets (groups of items, such as A and B, that satisfy a minimum support threshold, or percentage of task-related tuples), and then generating strong association rules in the form of A=>B.
- For frequent itemset mining, several powerful and scalable algorithms have been created, from which association and correlation rules can be extracted. These algorithms can be classified into three categories: (1) Apriori-like algorithms, (2) frequent pattern growth–based algorithms such as FP-growth.

Keywords

Antecedent: An antecedent is something that's found in data, and a consequent is an item that is found in combination with the antecedent.

Support: Support indicates how frequently the if/then relationship appears in the database.

Confidence: Confidence tells about the number of times these relationships are true.

Frequent Itemsets: The sets of an item that has minimum support

Apriori Property: Any subset of frequent item-set must be frequent.

Join Operation: To find Lk, a set of candidate k-item-sets is generated by joining Lk-1 with itself.

Sequence analysis algorithms: This type summarizes frequent sequences or episodes in data.

Association algorithms: This type of algorithm finds correlations between differentattributes in a dataset. The most common application of this kind of algorithm is for creating association rules, which can be used in a market analysis.

Self Assessment

1.A collection of one or more items is called as _____

A. Itemset
B. Support
C. Confidence
D. Support Count

2. Frequency of occurrence of an itemset is called as _____

A. Support
B. Confidence
C. Support Count
D. Rules

3. An itemset whose support is greater than or equal to a minimum support threshold is ______

A. (a)Itemset
B. (b)Frequent Itemset
C. (c)Infrequent items
D. (d)Threshold values

4. What does FP growth algorithm do?

A. It mines all frequent patterns through pruning rules with lesser support
B. It mines all frequent patterns through pruning rules with higher support
C. It mines all frequent patterns by constructing a FP tree
D. It mines all frequent patterns by constructing an itemsets

5. What do you mean by support(A)?

A. Total number of transactions containing A
B. Total Number of transactions not containing A
C. Number of transactions containing A / Total number of transactions
D. Number of transactions not containing A / Total number of transactions

6. Which of the following is the direct application of frequent itemset mining?

A. Social Network Analysis
B. Market Basket Analysis
C. Outlier Detection
D. Intrusion Detection

7. What is not true about FP growth algorithms?

A. It mines frequent itemsets without candidate generation
B. There are chances that FP trees may not fit in the memory
C. FP trees are very expensive to build
D. It expands the original database to build FP trees

8. When do you consider an association rule interesting?

A. If it only satisfies min_support
B. If it only satisfies min_confidence
C. If it satisfies both min_support and min_confidence
D. There are other measures to check so

9. What is the relation between a candidate and frequent itemsets?

A. A candidate itemset is always a frequent itemset

B. A frequent itemset must be a candidate itemset

C. No relation between these two

D. Strong relation with transactions

10. Which algorithm requires fewer scans of data?

A. Apriori

B. FP Growth

C. Naive Bayes

D. Decision Trees

11. For the question given below consider the data Transactions :

A. I1, I2, I3, I4, I5, I6

B. I7, I2, I3, I4, I5, I6

C. I1, I8, I4, I5

D. I1, I9, I10, I4, I6

E. I10, I2, I4, I11, I5

With support as 0.6 find all frequent itemsets?

A. <I1>, <I2>, <I4>, <I5>, <I6>, <I1, I4>, <I2, I4>, <I2, I5>, <I4, I5>, <I4, I6>, <I2, I4, I5>

B. <I2>, <I4>, <I5>, <I2, I4>, <I2, I5>, <I4, I5>, <I2, I4, I5>

C. <I11>, <I4>, <I5>, <I6>, <I1, I4>, <I5, I4>, <I11, I5>, <I4, I6>, <I2, I4, I5>

D. <I1>, <I4>, <I5>, <I6>

12. What is association rule mining?

A. Same as frequent itemset mining

B. Finding of strong association rules using frequent itemsets

C. Using association to analyze correlation rules

D. Finding Itemsets for future trends

13. The basic idea of the apriori algorithm is to generate________ item sets of a particular size & scans the database.

A. Primary.

B. Candidate.

C. Secondary.

D. Superkey.

14. This approach is best when we are interested in finding all possible interactions among a set of attributes.

A. Decision tree

B. Association rules

C. K-Means algorithm

D. Genetic learning

15. Which of the following is not a frequent pattern mining algorithm?

A. Apriori

B. FP growth

C. Decision trees

D. Eclat

Answer for Self Assessment

1.	A	2.	C	3.	B	4.	C	5.	C
6.	B	7.	D	8.	C	9.	B	10.	B
11.	A	12.	B	13.	B	14.	B	15.	C

Review Questions

Q1) The 'database' below has four transactions. What association rules can be found in this set, if the

minimum support (i.e coverage) is 60% and the minimum confidence (i.e. accuracy) is 80% ?

Trans_id Itemlist

T1 {K, A, D, B}

T2 {D, A C, E, B}

T3 {C, A, B, E}

T4 {B, A, D}

Q2) With appropriate example explain the concept of support and confidence in Association rule mining.

Q3) Discuss the various applications of Association rule learning.

Q4) Step-by-step exaplin the working of Apriori algorithm.

Q5) Differentiate between apriori and FP tree algorithm.

Q6) Elucidate the process how frequent items can be mined using FPtree.

Further Readings

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